AMATH 482: HOMEWORK 1

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Abstract. In this assignment, we determine the path of a submarine moving through the Puget Sound while emitting an unknown frequency. To do this, we must apply signal averaging and filtering in the Fourier domain to de-noise our recorded acoustic measurements. Specifically, we use a Gaussian filter around the dominant filter then return the de-noised signal to the spatial domain to plot the movement.

1. 1 Introduction and Overview

A submarine emitting an unknown frequency moves through the water and in this problem we attempt to locate it. To do this, we have noisy acoustic measurements recorded over 24 hours. We use the Fourier transform and a Gaussian filter in order to de-noise the measurements and see the submarine's path. This is visualized in Python to allow us to understand the submarine's movement.

2. 2 Theoretical Background

The basis of solving this problem is the Fourier transform, which converts the function in the spatial domain into the frequency domain by representing it as a series of sines and cosines. The transform is given by

$$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} f(x) dx$$

The function can be reverted back into the spatial domain by the inverse Fourier transform given by

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx} F(k) dk$$

[1] Due to the limitations of computational software, we use the fast Fourier transform (FFT) and inverse fast Fourier transform (IFFT). These are efficient algorithms for computing the discrete Fourier transform (DFT) and its inverse (IDFT). These formulas approximate the Fourier transform with a Riemann sum. To do this, the spatial domain is discretized and the following formula is applied:

$$F(k) = \frac{1}{N} \sum_{N=0}^{1} f(x_i) e^{-2\pi \frac{kx}{L}}$$

Or, for the inverse:

$$f(x) = \sum_{N=0}^{1} F(k) e^{2\pi \frac{kx}{L}}$$

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Using this method allows us to efficiently transform the data into the frequency space in which we can better filter the noisy data to create a clearer vision of the submarine's path. Once in the frequency space, we average the acoustic measurements across all time segments, which begins to filter out noise. To filter the frequency more completely, we applied a Gaussian filter. The filter is defined by the function

$$G(k_x, k_y, k_z) = e^{-\frac{1}{2\sigma^2}((k_x-k_x^*)^2+(k_y-k_y^*)^2+(k_z-k_z^*)^2}$$

[2] in which

$$k_x^*$$

represents the center value, or the index at which the dominant frequency is located in the average of frequency measurements at all recorded times. After applying the Gaussian. we can convert our filtered data back into the spatial domain in order to map the submarine's de-noised trajectory using the IFFT, as shown above.

3. 3 Algorithm Implementation and Development

3.1. **3.1 Setup.** To begin, we upload the data file containing the acoustic measurement data and establish the domain.The spatial domain consists of 64 equally spaced points between -10 and 10, inclusive, in all axes (x-, y-, and z-axes). The frequency domain ranges from -32 to 31, inclusive, and must be normalized by a factor of

$$\frac{2\pi}{2L}$$

because the frequency domain axis must range from

$$-2\pi$$

to

$$2\pi$$

. We also create meshgrids corresponding to each of these domains to be used for plotting in later visualizations.

3.2. **3.2 Time-Averaging.** Since we do not know the frequency being emitted by the submarine, we must first shift the measurements into the frequency domain and average the measurements at each time, which will diminish the noise as it is presumed to be mean zero white noise and allow us to determine the submarine's frequency. This averaging also serves as a preliminary filtering step through this reduction of noisy signals. To do this, we create a 64 x 64 x 64 array of zeros to serve as a sum of all measurements. Then, we create a loop inside of which we take a 64 x 64 x 64 array representing the acoustic measurements at one time and apply the fft and fftshift as the shifted data are more convenient for plotting and visualization. The shifted data are then added to the sum array. Once the loop has gone through all measurements, we divide by the number of measurements to find the average. Finally, we take the argmax of the average to find that the frequency signature of the submarine,

$$k^*$$

, is (39, 49, 10).

3.3. **3.3 Filtering and Plotting.** Next we complete the filtering of the noisy data in the frequency domain by applying a 3-dimensional Gaussian filter centered on our k-value found in the last step. We calculate the Gaussian values by putting our meshgrid values into the Gaussian filter, then use a loop to multiply the measurements at each time by these Gaussian values. Finally, we can convert back into the spatial domain by taking the fftshift again then the IIFT. We average the data in the spatial domain as we did with the data in the frequency domain in order to create a plot of the submarine's path. We normalize the data by dividing the absolute value of the averaged data by

the maximum value in this average before plotting it and seeing the submarine's movement over the recorded time.

4. 4 Computational Results

Our end result is a plot that shows the location of the submarine over time. The submarine's path begins on the left side and as we can see moves towards the right.

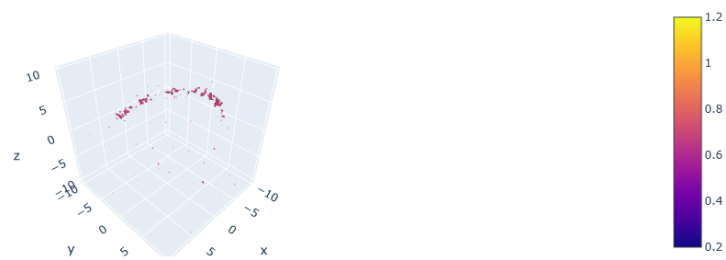

Figure 1. Submarine's path

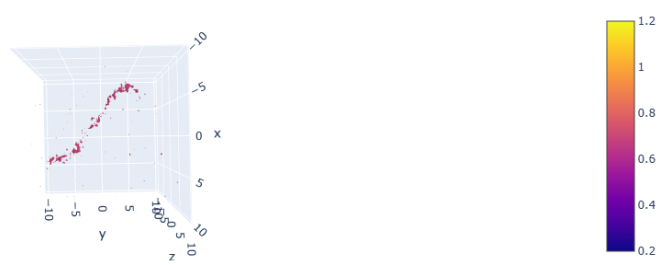

Figure 2. Top view of the submarine's path

As we can see in Figure 1, the submarine becomes slightly shallower over time. It also moves from the negative y-axis to the more positive part of the y-axis. However, the movement of the submarine on the x and y-axes is more clear in Figure 2. Here we can see that the submarine moves in a roughly diagonal path from the bottom left corner (negative y, positive x) to the top right corner (positive y, negative x).

5. 5 Summary and Conclusions

In summary, we used Fourier transform, averaging, and Gaussian filtering in order to find the path of a submarine emitting an unknown frequency as it moves through the Puget Sound. After applying this process, we discovered the path shown in Figure1 and Figure2 above. This information could be used to watch the submarine via a sub-tracking aircraft in the future.

6 Acknowledgments

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References

[1] Natalie Frank. Extensions of fourier series. AMATH 482, January 16, 2026.
[2] Natalie Frank. Fft and filtering. AMATH 482, January 16, 2026.